

Lattice non-perturbative foundation of generalized symmetry and applications

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- M. Abe, OM, S. Onoda, H. Suzuki and Y. Tanizaki, JHEP **08**, 118 (2023) [2303.10977].
- Y. Honda, OM, S. Onoda and H. Suzuki, PTEP **2024**, no.4, 043B04 (2024) [2401.01331].
- and many collaborated works

- 1 Introduction
 - Symmetry and 't Hooft anomaly
 - Generalized global symmetry
- 2 Review on topology of lattice gauge fields [Lüscher]
 - Principal fiber bundle
 - Construction of topology of lattice gauge fields
- 3 Fractionality of topology in lattice $SU(N)/\mathbb{Z}_N$ gauge theory [Abe–OM–Suzuki, Abe–OM–Onoda–Suzuki–Tanizaki]
- 4 Axial $U(1)$ non-invertible symmetry [Honda–Onoda–OM–Suzuki]
 - Lattice realization of axial $U(1)$ non-invertible symmetry
- 5 Summary

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Symmetry

- **Symmetry**: fundamental tool in physics
 - ▶ Universal applications to high energy physics, condensed matter physics and mathematics
- Noether theorem and conservation law

$$\text{Sym} : \phi \mapsto \phi'; S(\phi) = S(\phi'), \quad \partial_\mu j^\mu = 0$$

$$\text{Charge } Q \equiv \int j^0 d^{D-1}x$$

- Structure in the nature: Lorentz symmetry, *CPT* theorem, **Gauge invariance**, Internal symmetries

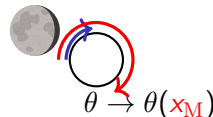
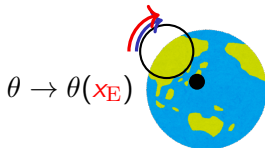
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- Structure in the nature: Lorentz symmetry, *CPT* theorem, **Gauge invariance**, Internal symmetries
 - ▶ Fundamental forces in the nature (e.g., photon: $U(1)$ sym)



't Hooft anomaly matching

- More aspects of Symmetry
 - ▶ Landau theory: phase transition (or vacuum structure)
 - ▶ Symmetry breaking: quantum anomaly, spontaneous breaking
- 't Hooft anomaly matching for strongly coupled theories [79]
 - ▶ Assume global symmetry G in system
 - ▶ Introduce background gauge field A assoc. G (by gauging G)

$$\begin{aligned} \mathcal{Z}[A] &= \int \mathcal{D}\{\phi\} e^{-S(\{\phi\}, A)} \\ &\stackrel{?}{=} \mathcal{Z}[A^g] = \dots = e^{\mathcal{A}[A, g]} \mathcal{Z}[A] \end{aligned}$$

$e^{\mathcal{A}} \neq 1 \xrightarrow{\text{Anomalous}}$ 't Hooft anomaly which is invariant at any energy scale (renormalization group inv.)

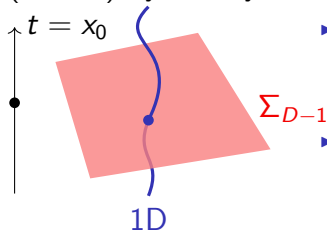
- ▶ Restriction on low-energy dynamics: SSB, phase structure, SPT

Recent generalization of symmetry

- **Generalized global symmetry** [Gaiotto–Kapustin–Seiberg–Willet '14]
 - ▶ Coupled with topological field theory
 - ★ Changing topological structure without changing local dynamics [Kapustin–Seiberg '14]:
 - ★ Nontrivial information by using generalized 't Hooft anomaly matching [E.g., OM–Wada–Yamaguchi '23]
 - Basic property: fractionality of topological charge
 - ▶ Usually, topo. charge $Q \sim \int F\tilde{F} \in \mathbb{Z}$ under topo. sectors
 - ★ θ term: θQ (strong CP problem, axion physics, sign problem)
 - ▶ **Discrete higher-form sym** $\rightarrow$ background gauge field B :
 $Q \sim \int B\tilde{B} \in \frac{1}{N}\mathbb{Z}$, 't Hooft anomaly $\mathcal{Z}_{\theta+2\pi}[B] = e^{-2\pi i Q} \mathcal{Z}_\theta[B]$
[Kapustin–Thorngren '13, Gaiotto–Kapustin–Komargodski–Seiberg '17]
- global desc.
 $\Rightarrow$ 't Hooft twisted boundary condition $Q \sim \int F\tilde{F} \in \frac{1}{N}\mathbb{Z}$
[van Baal '82] cf. [Edwards–Heller–Narayanan, de Forcrand–Jahn, Fodor–Holland–Kuti–Nógrádi–Schroeder, Kitano–Suyama–Yamada, Itou, ...]

Generalization: higher-form symmetry

- (0-form) Symmetry



- ▶ Charge (codim 1)

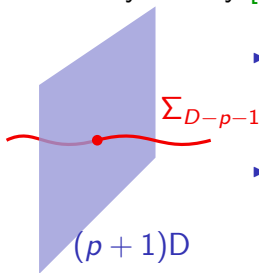
$$Q \equiv \int_{\Sigma_{D-1}} j_0 dx_1 \wedge \cdots \wedge dx_{D-1}$$

- ▶ Symmetry operator

$$U_\alpha(\Sigma_{D-1}) = e^{i\alpha Q}$$

Topological under deformation of Σ_{D-1}

- Higher-form symmetry [Gaiotto–Kapustin–Seiberg–Willet '14]



- ▶ p -form symmetry $G^{[p]}$ (codim $p+1$)

$$Q \equiv \int_{\Sigma_{D-p-1}} \star j^{(p+1)}, \quad U_\alpha(\Sigma) = e^{i\alpha Q}$$

- ▶ Transforming a “loop” operator $W(C)$

$$W(C) \mapsto U(\Sigma)W(C)$$

$$= e^{i\alpha \#(\Sigma, C)} W(C) \quad \text{w/ linking } \#$$

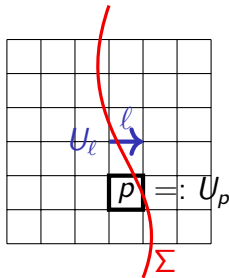
Center symmetry in YM theory

- Lattice $SU(N)$ YM theory
 - ▶ link variable $U_\ell \in SU(N)$

- Center symmetry: $\mathbb{Z}_N^{[1]}$

$$e^{\frac{2\pi i}{N}k} \in \mathbb{Z}_N \subset SU(N); \quad U_\ell \mapsto e^{\frac{2\pi i}{N}k \#(\Sigma, \ell)} U_\ell$$

Intersection $\#$ of Σ & link ℓ ; $U_p \mapsto U_p$

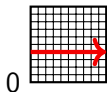


- Gauging the center symmetry

$$S \sim \sum \text{Tr} e^{-\frac{2\pi i}{N} B_p} U_p \quad B_p: \text{2-form gauge field assoc. } \mathbb{Z}_N^{[1]}$$

invariant under $U_\ell \mapsto e^{\frac{2\pi i}{N}\lambda_\ell} U_\ell$, $B_p \mapsto B_p + (d\lambda)_p$

- ▶ Recall 't Hooft twisted b.c. [79]: $U_{n+L\hat{\nu}, \mu} = g_{n, \nu}^{-1} U_{n, \mu} g_{n+\hat{\mu}, \nu}$



$$\text{gauge transf } g_{n+L\hat{\nu}, \mu}^{-1} g_{n, \nu}^{-1} g_{n, \mu} g_{n+L\hat{\mu}, \nu} = e^{\frac{2\pi i}{N} z_{\mu\nu}} \in \mathbb{Z}_N$$

$$\text{'t Hooft flux } z_{\mu\nu} = \sum B_p \bmod N$$

Our enterprise: Fully lattice regularized framework

- Wise but *not transparent* understanding

- ▶ Topological objects from **lattice** viewpoint as center sym

- ▶ Formal discussion in **continuum** theory

- ★ $\mathbb{Z}_N^{[q]}$ gauge field: **$U(1)$** field $B^{(q)}$

- with constraint $NB^{(q)} = dB^{(q-1)}$ from charge- N Higgs

- ▶ $Q \sim \frac{1}{N} \int B \wedge B? \xrightarrow[\text{cohomology op}]{\text{Swapping } \wedge \text{ w/}} \frac{1}{N} \int P_2(B) \sim -\frac{\varepsilon_{\mu\nu\rho\sigma} Z_{\mu\nu} Z_{\rho\sigma}}{8N}$

- ▶ Indicating mixed 't Hooft anomaly with chiral sym/ θ -periodicity

$$\mathcal{Z}_{\theta+2\pi}[B_p] = e^{-2\pi i Q} \mathcal{Z}_{\theta}[B_p] \quad \text{not } 2\pi \text{ periodic}$$

- **Fully regularized framework**: Lattice regularization

- ▶ Lattice construction of fiber bundle and $Q \in \mathbb{Z}$ [Lüscher '84]

- ▶ $Q \in \frac{1}{N}\mathbb{Z}$ [Abe-OM-Suzuki, Abe-OM-Onoda-Suzuki-Tanizaki]

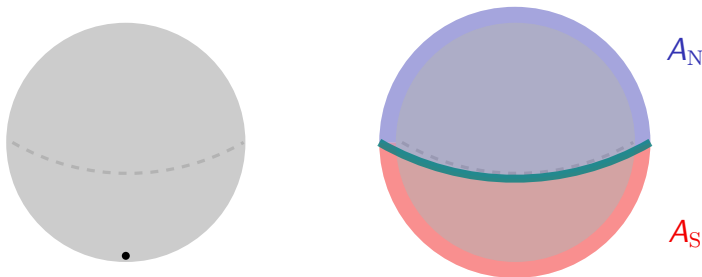
- ★ Higher-group [Kan-OM-Nagoya-Wada, Abe-OM-Onoda],
Magnetic operators [Abe-OM-Onoda-Suzuki-Tanizaki,
OM-Onoda-Suzuki]

- Application to **non-invertible symmetry**

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Fiber bundle and quantization

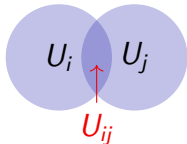
- Recall Dirac's discussion
 - ▶ Gauge fields cannot be defined globally in spacetime
- For instance, sphere:
 - ▶ Configuration on (θ, ϕ) is singular at $\theta = 0$ or π
 - ▶ Defined on each subspace: northern/southern hemisphere



- ▶ On **overlapping region**, $A_N - A_S = d\varphi$ (gauge)
- ▶ Due to consistency of wave function, $\int_{\text{equator}} \varphi \in 2\pi\mathbb{Z}$

Principal fiber bundle

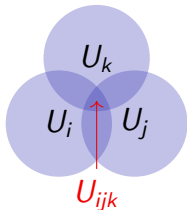
- Manifold (spacetime) X
- open covering (patches) $\{U_i\}$
- Gauge group G , gauge field a_i on U_i
- Relation between a_i and a_j ?



Gauge transformation:

$$a_j = g_{ij}^{-1} a_i g_{ij} + g_{ij}^{-1} d g_{ij} \quad \text{at } U_{ij}$$

- Consistency condition for transition function g_{ij} :

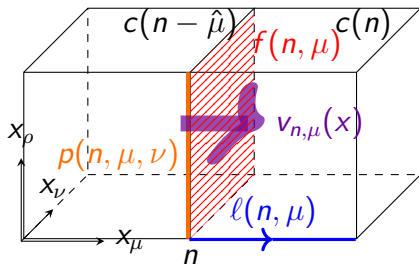


$$g_{ii} = \text{id}, \quad g_{ji} = g_{ij}^{-1}$$

Cocycle condition: $g_{ij} g_{jk} g_{ki} = 1 \quad \text{at } U_{ijk}$

Bundle structure on lattice?

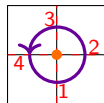
- No continuity for lattice fields?
 - ▶ Any configuration can be deformed continuously to others
 - ▶ We can observe topological structure even on lattice by Lüscher
- Setup/strategy: Lattice Λ divides X into 4D hypercubes



4D cell $c(n)$
 3D face $f(n, \mu)$
 2D plaquette $p(n, \mu, \nu)$
 1D link $\ell(n, \mu)$

- 1 Regard $\{c(n)\}$ as patches
- 2 Define transition function $v_{n, \mu}(x)$ at $f(n, \mu)$ from data as U_ℓ

- ▶ Difficult to define it at $x \neq n$ s.t.
cocycle condition is kept intact



$$v_1 v_2 v_3^{-1} v_4^{-1} = 1$$

at $x \in p(n)$

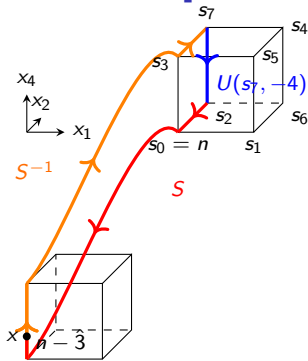
Construction of topo. sectors on lattice [Lüscher]

- Parallel transporter (interpolation):

$$U_\ell \rightarrow v_f(n) \rightarrow v_f(x) \Leftarrow \text{Cocycle}$$

$$v_{n,\mu}(n) = U(n - \hat{\mu}, \mu)$$

$$v_{n,\mu}(x) \equiv S_{n,\mu}^{n-\hat{\mu}}(x)^{-1} v_{n,\mu}(n) S_{n,\mu}^n(x)$$



- Topo. sectors on lattice so that $[Q \text{ in terms of } v_f(x)] \in \mathbb{Z}$

$$Q = \sum_{n \in \Lambda} q(n), \quad q(n) = -\frac{1}{24\pi^2} \sum_{\mu, \nu, \rho, \sigma} \epsilon_{\mu\nu\rho\sigma} \left\{ 3 \int_{\rho_{n,\mu\nu}} d^2x \, \text{Tr} \left[(v_{n,\mu} \partial_\rho v_{n,\mu}^{-1}) (v_{n-\hat{\mu},\nu}^{-1} \partial_\sigma v_{n-\hat{\mu},\nu}) \right] \right. \\ \left. + \int_{f_{n,\mu}} d^3x \, \text{Tr} \left[(v_{n,\mu}^{-1} \partial_\nu v_{n,\mu}) (v_{n,\mu}^{-1} \partial_\rho v_{n,\mu}) (v_{n,\mu}^{-1} \partial_\sigma v_{n,\mu}) \right] \right\}$$

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$$w^n(x) = U_{n,4}^{y_4} U_{n+y_4\hat{4},3}^{y_3} U_{n+y_4\hat{4}+y_3\hat{3},2}^{y_2} U_{n+y_4\hat{4}+y_3\hat{3}+y_2\hat{2},1}^{y_1}$$

$$f_{n,\mu}^m(x_\gamma) = (u_{s_3 s_0}^m)^{y_\gamma} (u_{s_0 s_3}^m u_{s_3 s_7}^m u_{s_7 s_2}^m u_{s_2 s_0}^m)^{y_\gamma} u_{s_0 s_2}^m (u_{s_2 s_7}^m)^{y_\gamma},$$

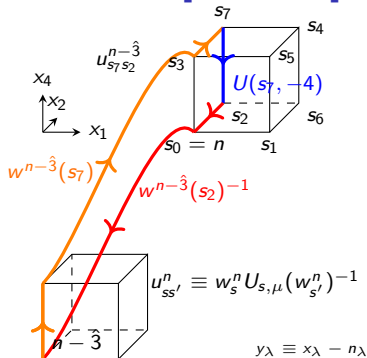
$$g_{n,\mu}^m(x_\gamma) = (u_{s_5 s_1}^m)^{y_\gamma} (u_{s_1 s_5}^m u_{s_5 s_4}^m u_{s_4 s_6}^m u_{s_6 s_1}^m)^{y_\gamma} u_{s_1 s_6}^m (u_{s_6 s_4}^m)^{y_\gamma},$$

$$h_{n,\mu}^m(x_\gamma) = (u_{s_3 s_0}^m)^{y_\gamma} (u_{s_0 s_3}^m u_{s_3 s_5}^m u_{s_5 s_1}^m u_{s_1 s_0}^m)^{y_\gamma} u_{s_0 s_1}^m (u_{s_1 s_5}^m)^{y_\gamma},$$

$$k_{n,\mu}^m(x_\gamma) = (u_{s_7 s_2}^m)^{y_\gamma} (u_{s_2 s_7}^m u_{s_7 s_4}^m u_{s_4 s_6}^m u_{s_6 s_2}^m)^{y_\gamma} u_{s_2 s_6}^m (u_{s_6 s_4}^m)^{y_\gamma},$$

$$l_{n,\mu}^m(x_\beta, x_\gamma) = [f_{n,\mu}^m(x_\gamma)^{-1}]^{y_\beta} [f_{n,\mu}^m(x_\gamma) k_{n,\mu}^m(x_\gamma) g_{n,\mu}^m(x_\gamma)^{-1} h_{n,\mu}^m(x_\gamma)^{-1}]^{y_\beta} h_{n,\mu}^m(x_\gamma) [g_{n,\mu}^m(x_\gamma)]^{y_\beta},$$

$$S_{n,\mu}^m(x_\alpha, x_\beta, x_\gamma) = (u_{s_0 s_3}^m)^{y_\gamma} [f_{n,\mu}^m(x_\gamma)]^{y_\beta} [l_{n,\mu}^m(x_\beta, x_\gamma)]^{y_\alpha}.$$



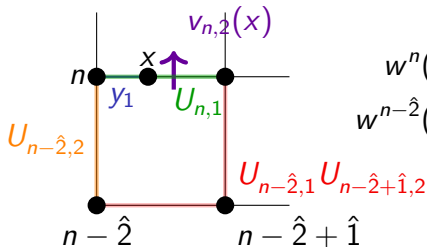
$$y_\lambda \equiv x_\lambda - n_\lambda$$

- Topo. sectors on lattice so that $[Q \text{ in terms of } v_f(x)] \in \mathbb{Z}$

$$Q = \sum_{n \in \Lambda} q(n), \quad q(n) = -\frac{1}{24\pi^2} \sum_{\mu, \nu, \rho, \sigma} \epsilon_{\mu\nu\rho\sigma} \left\{ 3 \int_{\rho_{n,\mu\nu}} d^2x \text{Tr} \left[(v_{n,\mu} \partial_\rho v_{n,\mu}^{-1}) (v_{n-\hat{\mu},\nu}^{-1} \partial_\sigma v_{n-\hat{\mu},\nu}) \right] \right. \\ \left. + \int_{f_{n,\mu}} d^3x \text{Tr} \left[(v_{n,\mu}^{-1} \partial_\nu v_{n,\mu}) (v_{n,\mu}^{-1} \partial_\rho v_{n,\mu}) (v_{n,\mu}^{-1} \partial_\sigma v_{n,\mu}) \right] \right\}$$

Exercise: Lattice 2D $U(1)$ gauge theory

- $v_{n,\mu}(x) = w^{n-\hat{\mu}}(x) w^n(x)^{-1}$ [Lüscher '98, Fujiwara et al. '00]



$$w^n(x) = U_{n,1}^{y_1}$$

$$w^{n-\hat{2}}(x) = [U_{n-\hat{2},1} U_{n-\hat{2}+\hat{1},2}]^{y_1} U_{n-\hat{2},2}^{1-y_1}$$

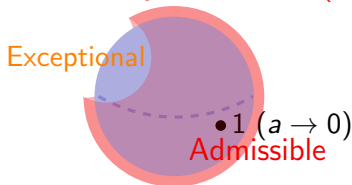
- Explicit expression of v :

$$\begin{aligned} v_{n,1}(x) &= U_{n-\hat{1},1} \quad v_{n,2}(x) = U_{n-\hat{2},2} [U_{n-\hat{2},1} U_{n-\hat{2}+\hat{1},2} U_{n,1}^{-1} U_{n-\hat{2},2}^{-1}]^{y_1} \\ &= U_{n-\hat{2},2} \exp[iy_1 F_{12}(n - \hat{2})] \end{aligned}$$

- ▶ Field strength: $F_p \equiv \frac{1}{i} \ln U_p$ for $-\pi < F_p \leq \pi$
- ▶ (nD) To ensure Bianchi identity $dF_p = 0$, we should impose $\sup_p |F_p| < \epsilon$, $0 < \epsilon < \frac{\pi}{3} \rightarrow$ **Admissibility condition** (see also §7)

Admissibility condition

- In general, admissibility = well-defined-ness of u^y ($0 \leq y \leq 1$)
 - ★ $U(1)$: $F_p = \frac{1}{i} \ln U_p$ for plaquette U_p
 - ★ $S_{n,\mu}^m(x)$ is written in terms of $(u_{ss'}^n)^y$ where u is a loop
 $n \rightarrow s \rightarrow s' \rightarrow n$
 - ▶ E.g., u^y is ill-defined at $u = -1$; ill-def regions separate **sectors**
- **Admissibility condition** $\text{tr}(1 - U_p) < \epsilon$ [Lüscher '84]



- ▶ Admissible lattice gauge fields: well-defined conf space $\sim$ disk
 - ▶ Exceptional region
 - ★ Topological freezing
 - ★ Monopole as lattice artifact
- Under the admissibility condition, we can prove that $Q \in \mathbb{Z}$; we observe topo. sectors even on lattice!
- How about index theorem for finite a ?

$$\text{Index}(D) = \underbrace{-\frac{a}{2} \text{Tr} \gamma_5 D_{\text{ov}}}_{\text{Admissibility } \epsilon_{\text{ov}}} = \underbrace{n_+ - n_-}_{\in \mathbb{Z}} \stackrel{?}{=} \frac{1}{32\pi^2} \int_x \varepsilon_{\mu\nu\rho\sigma} \text{tr}[F_{\mu\nu} F_{\rho\sigma}]$$

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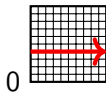
Cocycle condition relaxed by higher-form sym

- Lüscher's construction *Coupled* with higher-form gauge fields
 - ▶ $SU(N)$ YM theory coupled with $\mathbb{Z}_N$ 2-form gauge field

$$S \sim \sum \text{Tr} e^{-\frac{2\pi i}{N} B_p} U_p \quad B_p: \text{2-form gauge field assoc. } \mathbb{Z}_N^{[1]}$$

invariant under $U_\ell \mapsto e^{\frac{2\pi i}{N} \lambda_\ell} U_\ell$, $B_p \mapsto B_p + (d\lambda)_p$

- ▶ 't Hooft twisted b.c. ['79]: $U_{n+L\hat{\nu},\mu} = g_{n,\nu}^{-1} U_{n,\mu} g_{n+\hat{\mu},\nu}$



$$\text{gauge transf } g_{n+L\hat{\nu},\mu}^{-1} g_{n,\nu}^{-1} g_{n,\mu} g_{n+L\hat{\mu},\nu} = e^{\frac{2\pi i}{N} z_{\mu\nu}} \in \mathbb{Z}_N$$

$$\text{'t Hooft flux } z_{\mu\nu} = \sum B_p \bmod N$$

- Cocycle condition can take a $\mathbb{Z}_N$ value:

$$\tilde{v}_{n-\hat{\nu},\mu}(x) \tilde{v}_{n,\nu}(x) \tilde{v}_{n,\mu}(x)^{-1} \tilde{v}_{n-\hat{\mu},\nu}(x)^{-1} = e^{\frac{2\pi i}{N} B_{\mu\nu}(n-\hat{\mu}-\hat{\nu})}$$

- ▶ $\mathbb{Z}_N$ blind matters: adjoint repr.
- ▶ $\mathbb{Z}_N^{[1]}$ gauge inv. if $\tilde{v}_{n,\mu}(x) \mapsto e^{\frac{2\pi i}{N} \lambda_{n-\hat{\mu},\mu}} \tilde{v}_{n,\mu}(x)$
- What is definition of $\tilde{v}_{n,\mu}(x)$?

$\mathbb{Z}_N^{[1]}$ gauge invariant construction

- Gauge invariant plaquette

$$\tilde{U}_p \equiv e^{-\frac{2\pi i}{N} B_p} U_p$$

- Recall $U_\ell \mapsto e^{\frac{2\pi i}{N} \lambda_\ell} U_\ell$
- Admissibility $\text{tr}(1 - \tilde{U}_p) < \epsilon$

- u : product of plaquettes $\rightarrow \tilde{u}$

$$\tilde{u}_{s_7 s_2}^{n-\hat{3}} = e^{\frac{2\pi i}{N} B_{34}(n-\hat{3})} e^{\frac{2\pi i}{N} B_{24}(n)} u_{s_7 s_2}^{n-\hat{3}}$$

- Similarly, $\tilde{v}$ is defined in terms of $\tilde{u}$

- Gauge covariance

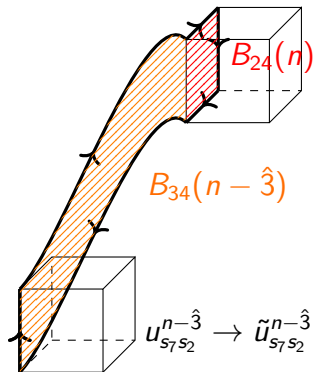
$$\tilde{v}_{n,\mu}(x) \mapsto e^{\frac{2\pi i}{N} \lambda_\mu(n-\hat{\mu})} \tilde{v}_{n,\mu}(x)$$

- Fractional topological charge $Q = \sum_n q(n) \in -\frac{\varepsilon_{\mu\nu\rho\sigma} Z_{\mu\nu} Z_{\rho\sigma}}{8N} + \mathbb{Z}$

$$q(n) = -\frac{1}{8N} \sum_{\mu,\nu,\rho,\sigma} \varepsilon_{\mu\nu\rho\sigma} B_{\mu\nu}(n) B_{\rho\sigma}(n + \hat{\mu} + \hat{\nu}) + \check{q}(n)$$

where $\sum_n \check{q}(n) \in \mathbb{Z}$ (each term in q is not inv. under $\mathbb{Z}_N^{[1]}$)

- We observe mixed 't Hooft anomaly with lattice regularization!



Numerical simulations of higher-form gauge fields

Public repository: <https://github.com/o-morikawa/Gaugefields.jl>

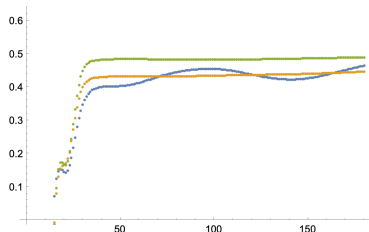
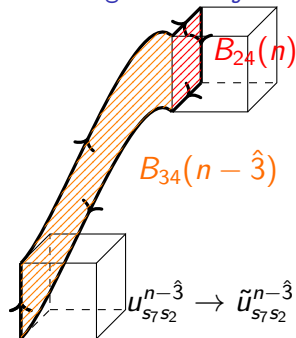
- Obviously, we can implement it on computer for numerical simulations
- $\forall u$, we have 1-form gauge invariant $\tilde{u}$

$$\tilde{u}_{s_7 s_2}^{n-\hat{3}} = e^{\frac{2\pi i}{N} B_{34}(n-\hat{3})} e^{\frac{2\pi i}{N} B_{24}(n)} u_{s_7 s_2}^{n-\hat{3}}$$

even if missing a link (covariant)

- External/dynamical B to carry out HMC, gradient flow and MPI
- You can see immediately fractional topo. charge
 - ▶ $SU(2)$: Green points (lattice improved Q) go to $1/2!!!$

- [on-going Abe-OM-Suzuki,
Kanamori-OM-Tanizaki-Tomiya-Watanabe]



- 1 Introduction
 - Symmetry and 't Hooft anomaly
 - Generalized global symmetry
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 - Principal fiber bundle
 - Construction of topology of lattice gauge fields
- 3 Fractionality of topology in lattice $SU(N)/\mathbb{Z}_N$ gauge theory [Abe-OM-Suzuki, Abe-OM-Onoda-Suzuki-Tanizaki]
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Generalized symmetry: Non-invertibility

LIE ALGEBRAS IN

- PARTICLE PHYSICS by H. Georgi

“Group theory is the study of symmetry.”

- Based on recent developments of generalized symmetry, symmetry is not necessarily described by group!

- 1 Associativity: $(ab)c = a(bc)$ for $\forall a, b, c$
- 2 Identity element: $\exists e$ s.t. $ae = ea = a$
- 3 **Inverse** element: $\exists b$ s.t. $ab = ba = e$ ($b = a^{-1}$)

So far, symmetry possesses invertibility.

- Now, for “naively unitary” symmetry operators $\{U_\alpha = e^{i\alpha Q}\}$, there can exist **Non-invertible symmetry** in some systems

$$\mathcal{D} \times \mathcal{D}^\dagger \neq 1$$

[Many studies; See recent lectures by Schäfer-Nameki, Shao]

- Let's consider non-invertible symmetry for axial $U(1)$ transf, and then, in this talk, our aim is to realize it on lattice!

Axial $U(1)$ symmetry in continuum theory

- Quantum anomaly $\rightarrow$ loss of symmetry
ABJ anomaly: no conserved current for axial $U(1)$ symmetry
- For fractional rotation angles, conserved, gauge-invariant and topological **non-invertible** symmetry exists
[Córdova–Ohmori '22, Choi–Lam–Shao '22]
- Naive symmetry operator

$$d \star j = \frac{1}{4\pi^2} F \wedge F, \quad U_\alpha(M) = \exp \left(\frac{i\alpha}{2} \int_M \star j \right)$$

is not topological

- Let us consider topological modification by Chern–Simons:

$$\hat{U}_\alpha(M) = \exp \left[\frac{i\alpha}{2} \int \left(\star j - \frac{1}{4\pi^2} A \wedge dA \right) \right]$$

But gauge non-invariant!!

- For $\alpha = \frac{2\pi}{N}$, $-i \int_M \frac{1}{4\pi N} A \wedge dA$ is still gauge non-invariant because of fractional CS level $1/N$

Fractional quantum Hall effect and non-invertibility

- TQFT (fractional quantum Hall state) as

$$i \int_M \left(\frac{N}{4\pi} a \wedge da + \frac{1}{2\pi} a \wedge dA \right)$$

which is gauge-invariant with fractional CS level

- ▶ a : additional dynamical $U(1)$ gauge field
- ▶ Naively, this action is (level- N CS) + $(A_\mu J^\mu)$; $a = -A/N$
“Substituting” this into action reproduces $-i \int \frac{1}{4\pi N} A \wedge dA$
- To modify naive symmetry operator $U_\alpha(M)$, we use this action at boundary M instead of naive CS term:

$$\mathcal{D}_{1/N}(M) = \exp \left[i \int \left(\frac{\pi}{N} \star j + \frac{N}{4\pi} a \wedge da + \frac{1}{2\pi} a \wedge dA \right) \right]$$

- Gauge invariant and topological; no inverse operator

$$\mathcal{D}_{1/N} \times \mathcal{D}_{1/N}^\dagger = \mathcal{C} \neq 1$$

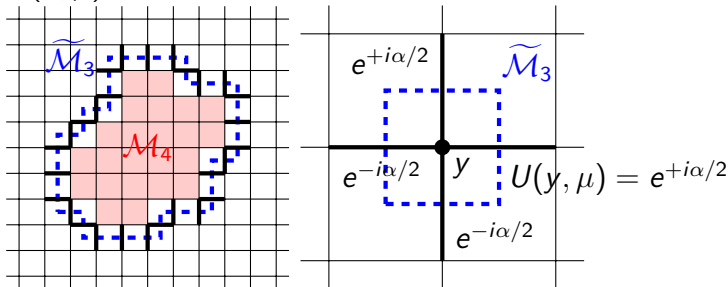
- ▶ $\mathcal{C}$: condensation operator

Lattice realization?

- We want to realize lattice axial $U(1)$ non-invertible symmetry
- Problems:
 - ▶ Lattice Chern–Simons theory?
 - ★ cf. Villain formulation of $U(1)$ GT [Jacobson–Sulejmanpasic '23]
 $\nrightarrow$ non-Abelian generalization
 - ▶ (Anomaly-free) Chiral lattice gauge theory?
 - ★ For $U(1)$ gauge theory, Lüscher's construction ['98]
 - ▶ Anomalous gauge theory? (Continuation of studying it?)
 - ★ E.g., [Forster–Nielsen–Ninomiya '80, Harada–Tsutsui '87, ...]
 - ★ Lattice framework: no-go as in [Kikukawa–Suzuki '07]
- Construct (fermion measure in) **anomalous** chiral gauge theory
 - ▶ Introduce external $U(1)'$ gauge group
 - 1 to manipulate axial rotation (Chiral $U(1)' \ni U(x, \mu)$)
 - 2 to define topological defect
 - 3 to keep locality for **dynamical** fields (Vector $U(1) \ni u(x, \mu)$)
based on Lüscher's construction (§8)

Introduction of topological defect

- $U(x, \mu) = e^{\pm i\alpha/2}$ for 3D closed surface $\widetilde{\mathcal{M}}_3$; otherwise 1



- Anomalous Ward–Takahashi identity is

$$\langle \mathcal{O} \rangle_{\text{F}}^{\widetilde{\mathcal{M}}_3} = \exp \left[-i\alpha\gamma \sum_{x \in \mathcal{M}_4} \epsilon_{\mu\nu\rho\sigma} f_{\mu\nu}(x) f_{\rho\sigma}(x + \hat{\mu} + \hat{\nu}) \right] \langle \mathcal{O}^\alpha \rangle_{\text{F}},$$

where $\psi(x)^\alpha = e^{-i\alpha/2}\psi(x)$, $\bar{\psi}(x)^\alpha = \bar{\psi}(x)e^{i\alpha/2}$ for $x \in \mathcal{M}_4$

- Symmetry operator may be written by

$$\left\langle U_\alpha(\widetilde{\mathcal{M}}_3) \mathcal{O} \right\rangle_{\text{F}} := \langle \mathcal{O} \rangle_{\text{F}}^{\widetilde{\mathcal{M}}_3} \exp \left[i\alpha\gamma \sum_{x \in \mathcal{M}_4} \epsilon_{\mu\nu\rho\sigma} f_{\mu\nu}(x) f_{\rho\sigma}(x + \hat{\mu} + \hat{\nu}) \right]$$

Coupling with $\mathbb{Z}_N$ TQFT for $\alpha \sim 2\pi/N$

- Hard problem: lattice Chern–Simons with rotation angle α
- Instead of CS, **lattice $\mathbb{Z}_N$ TQFT** (thanks to Tanizaki-san)
- Set rotation angles α as $2\pi p/N$ ($p, N \in \mathbb{Z}$)
- It is natural to consider 3D level- N BF theory on $\mathcal{M}_3$

$$S_{\text{BF}} = -\frac{ip\pi}{N} \sum_{x \in \mathcal{M}_3} \epsilon_{\mu\nu\rho} \left\{ b_\mu(\tilde{x}) \left[\partial_\nu c(x + \hat{\mu}) - \frac{1}{2} z_{\nu\rho}(x + \hat{\mu}) \right] \right. \\ \left. - \frac{1}{2} z_{\mu\nu}(x) c_\rho(x + \hat{\mu} + \hat{\nu}) \right\}$$

► Dual lattice $\tilde{x}$; $f = \delta a + 2\pi z$, $-\pi < a_\mu(x) \equiv \frac{1}{i} \ln u(x, \mu) \leq \pi$

- For simplicity, $S_{\text{BF}} = \frac{-ip\pi}{N} \sum [b(\delta c - z) - z \cup c]$

Symmetry operator in terms of $\mathbb{Z}_N$ TQFT

- Consider $\mathcal{Z}_{\mathcal{M}_3}[z] = \frac{1}{N^s} \int DbDc e^{-S_{\text{BF}}}$, s : # of sites $\in \mathcal{M}_3$
- From summation over b , $\mathcal{Z}_{\mathcal{M}_3}[z] = 0$ if $\sum_{\mathcal{M}_2} z \neq 0 \bmod N$
 - ▶ If $z = 0$, $\mathcal{Z}_{\mathcal{M}_3}[0] = N^{b_2-1}$, where b_2 : 2nd Betti number of $\mathcal{M}_3$
 - ▶ If $z = \delta\nu \bmod N$,

$$\mathcal{Z}_{\mathcal{M}_3}[z] = \exp \left(-\frac{ip\pi}{N} \sum_{\mathcal{M}_3} \delta\nu \cup \nu \right) \mathcal{Z}_{\mathcal{M}_3}[0]$$

- Redefine symmetry operator on arbitrary 3-cycle $\widetilde{\mathcal{M}}_3$:

$$\left\langle \tilde{U}_{\frac{2\pi p}{N}}(\widetilde{\mathcal{M}}_3) \mathcal{O} \right\rangle_{\text{F}} = \langle \mathcal{O} \rangle_{\text{F}}^{\widetilde{\mathcal{M}}_3} \exp \left[-\frac{ip}{4\pi N} \sum_{\mathcal{M}_3} (a \cup f + 2\pi z \cup a) \right] \mathcal{Z}_{\mathcal{M}_3}[z]$$

- ▶ Then, topological: $\left\langle \tilde{U}_{\frac{2\pi p}{N}}(\widetilde{\mathcal{M}}'_3) \mathcal{O} \right\rangle_{\text{F}} = \left\langle \tilde{U}_{\frac{2\pi p}{N}}(\widetilde{\mathcal{M}}_3) \mathcal{O} \right\rangle_{\text{F}}$
- ▶ Gauge invariant: cancellation between exp and $\mathcal{Z}_{\mathcal{M}_3}$ under $a \mapsto a - \delta\phi - 2\pi l$ and $z \mapsto z + \delta l$

Fusion rules

- From $\mathcal{Z}_{\mathcal{M}_3}^{(p,N)}[z]\mathcal{Z}_{-\mathcal{M}_3}^{(p,N)}[z] = \mathcal{Z}_{\mathcal{M}_3}[0]\mathcal{C}_{\mathcal{M}_3}[z]$, condensation operator:

$$\mathcal{C}_{\mathcal{M}_3}[z] = \frac{1}{N^s} \int DbDc e^{b(\delta c - z)}$$

- For different p_1 and p_2 (assuming $\gcd(p_1 + p_2, N) = 1$)

$$\tilde{U}_{2\pi p_1/N} \tilde{U}_{2\pi p_2/N} = \mathcal{Z}_{\mathcal{M}_3}[0] \tilde{U}_{2\pi(p_1+p_2)/N}$$

- More generally ($\gcd(p_1, N_1) = 1$, $\gcd(p_2, N_2) = 1$, $\gcd(N[p_1/N_1 + p_2/N_2], N) = 1$ with $N = \text{lcm}(N_1, N_2)$),

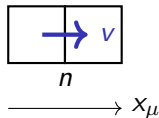
$$\tilde{U}_{2\pi p_1/N_1} \tilde{U}_{2\pi p_2/N_2} = \frac{\mathcal{Z}_{\mathcal{M}_3}^{(N_1)}[0]\mathcal{Z}_{\mathcal{M}_3}^{(N_2)}[0]}{\mathcal{Z}_{\mathcal{M}_3}^{(N)}[0]} \tilde{U}_{2\pi(p_1/N_1 + p_2/N_2)}$$

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Summary [1/3]

- Generalized symmetries have been developed in this decade
 - Higher-form sym, higher-group sym, noninvertible sym, ...
 - Through the use of 't Hooft anomaly matching, new insights about *nontrivial dynamics & classification of phases*
- Standing on a **fully regularized framework: lattice gauge theory**
 - Generalization of Lüscher's construction of topology on lattice
 - Maintaining **locality**, **$SU(N)$ gauge inv** & **higher-form gauge inv**
 - There exists interpolation to smooth enough bundle structure

- ★ Transition function $v_f(n) \rightarrow v_f(x)$
- ★ Q is written in terms of $v_f(x)^{-1} \partial_\nu v_f(x)$



$$Q \in \mathbb{Z} \xrightarrow{\text{Gauging } \mathbb{Z}_N^{[1]}} \frac{1}{N} \mathbb{Z}$$

- ▶ Mixed 't Hooft anomaly between $\mathbb{Z}_N^{[1]}$ & θ periodicity

Summary [2/3]

- Some simple applications

- ▶ **Higher-group** symmetry from lattice (§6)
 - ★ Mixture of different higher-form symmetries
- ▶ Lattice construction of **magnetic operators** and observation of **Witten effect** (§7)
 - ★ Admissibility condition implies absence of monopole
 - ★ Our proposal: Excision method (opening a hole in lattice)
 - ★ By arranging lattice/dual-lattice in 2D scalar theory, mixing magnetic charges with electric ones (Witten effect)
- ▶ Future works
 - ★ Construction of monopole, 't Hooft line in gauge theory [See e.g., Honda–Onoda–Suzuki '24, OM–Onoda–Suzuki '24]
 - ★ Numerical simulation of $SU(N)$ gauge theory coupled with $\mathbb{Z}_N$ 2-form gauge fields [on-going Abe–OM–Suzuki, Kanamori–OM–Tanizaki–Tomiya–Watanabe]

Summary [3/3]

- (Again) Standing on a **completely lattice regularized framework**
 - ▶ Generalized Lüscher's construction of chiral LGT (§8)
 - ▶ Construction of fermion measure: for dynamical fields, smooth and local; for external $U(x, \mu)$, **non-local** (unphysical)
- **Level- N BF theory** (thanks to Tanizaki-san)
 - ▶ We can construct symmetry operator for rational angles, and evaluate **fusion rules**!
 - ▶ Other attempt to Karasik's prescription (§9)
 - ★ Consider “gauge average” [’22]; there are some subtleties in our lattice formulation. . .
- Questions
 - ▶ Physical phenomena (e.g., monopole) [on-going Honda–Kan–OM]
 - ▶ Generalization to non-Abelian gauge theory
 - ▶ Some aspects of anomalous/chiral lattice gauge theory (w/ boundary)

- 6 Higher-group structure in lattice gauge theory
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Generalization: higher-group structure

- In general suppose $\otimes_{i,p} G_i^{[p]}$ global symmetry
- After gauging, a naive direct product of symmetry groups?
 - ▶ Can each symmetry be gauged *individually*?
- Gauging $G^{[0]} \times H^{[1]}$ global symmetry, then gauge transf.:

$$A \mapsto A + d\lambda^{(0)}, \quad B \mapsto B + d\lambda^{(1)} + \textcolor{red}{Ad}\lambda^{(0)}$$

2-group symmetry (cf. superstring theory [Green–Schwarz '84])

- ▶ p -group symmetry: $G_0^{[0]} \textcolor{red}{\tilde{\times}} \dots \textcolor{red}{\tilde{\times}} G_{p-1}^{[p-1]}$
- E.g., 4D $SU(N)$ gauge theory with instanton number $p\mathbb{Z}$
 - ▶ For any $p \in \mathbb{Z}$, local and unitary [Seiberg '10]
 - ▶ Global symmetry: $\underbrace{\mathbb{Z}_N^{\text{center sym}} \times \mathbb{Z}_p^{[3]} \text{ sym}}_{\text{gauging}} \xrightarrow{\text{gauging}} \text{4-group}$

[Tanizaki–Ünsal '19]

cf. [Hidaka–Nitta–Yokokura '21]
- How to modify instanton sum & realize higher-group on lattice?

Modified instanton-sum: higher-group symmetry

- For $Q = \sum_n q_n$, insert the delta function

$$\delta(q_n - pc_n) \rightarrow Q = \textcolor{red}{p} \sum_n c_n \in p\mathbb{Z}$$

where $U(1)$ 4-form field strength c_n

- Obviously no nontrivial configurations for B_p
- Introducing new field Ω_n ($\Omega_n \in \mathbb{R}$ and $\sum_n \Omega_n \in \mathbb{Z}$)
 - Replacement: $c_n \rightarrow c_n - \frac{1}{Np}\Omega_n$: 3-form gauge inv

$$q_n - pc_n + \frac{1}{N}\Omega_n = 0 \quad : \text{fractionality allowed}$$

- Redefine Ω_n as $\tilde{\Omega}_n \equiv \frac{1}{N}\Omega_n - \underbrace{\frac{1}{8N}\varepsilon_{\mu\nu\rho\sigma}B_{n,\mu\nu}B_{n+\hat{\mu}+\hat{\nu},\rho\sigma}}_{\text{fractional part of } Q}$

$$\text{Again } \sum_n \tilde{\Omega}_n \in \mathbb{Z}$$

$$\check{q}_n - pc_n + \tilde{\Omega}_n = 0 \quad \text{where } \check{q}_n: \text{integral part of } Q$$

- 1-form and 3-form gauge transf with $\Omega_n^{(3)} \in \mathbb{R}$:

$$B_p \mapsto B_p + (d\lambda)_p, \quad c_n \mapsto c_n + \frac{1}{p}d\Omega_n^{(3)} (+\mathbb{Z}),$$

$$\tilde{\Omega}_n \mapsto \tilde{\Omega}_n + d\Omega_n^{(3)} (+p\mathbb{Z}) + \textcolor{red}{\left[\frac{2}{N}B \wedge d\lambda + \frac{1}{N}d\lambda \wedge d\lambda \right]} (+\mathbb{Z})$$

- Mixture of 1-form and 3-form gauge transf

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Mixed 't Hooft anomaly in $SU(N)$ or $U(1)$ GT

- 1-form inv. action: $S_{\text{YM}} + S_{\text{matter}} - i\theta Q$

$$\mathcal{Z}_{\theta+2\pi}[B_p] = e^{-2\pi i Q[B_p]} \mathcal{Z}_{\theta}[B_p]$$

Anomaly between $\mathbb{Z}_N^{[1]}$ gauge inv. and θ periodicity

- In case of $U(1)$ gauge theory, gauging $\mathbb{Z}_q^{[1]} \subset U(1)_E$ symmetry

$$Q \in \frac{1}{8q^2} \varepsilon_{\mu\nu\rho\sigma} z_{\mu\nu} z_{\rho\sigma} + \frac{1}{4q} (\varepsilon_{\mu\nu\rho\sigma} z_{\mu\nu})_{\mu\nu \leftrightarrow \rho\sigma} \mathbb{Z} + \mathbb{Z}$$

Modified as $-iq\theta Q$ by Witten effect [Honda–Tanizaki '20]

- ▶ If $-i\check{\theta}Q$, $\check{\theta} \sim \check{\theta} + 2\pi q$; $\check{\theta} = q\theta$, then $\theta \sim \theta + 2\pi$
- ▶ Electric charge of dyon $= qe + \frac{1}{2\pi}\check{\theta}$ (matters with charge $q\mathbb{Z}$)
- ▶ 't Hooft anomaly $= e^{-2\pi i q Q}$ with $qQ \in \frac{1}{q}\mathbb{Z}$
- ▶ Under some kind of modification (see next §), there exist $\mathbb{Z}_N^{[1]}$ if multiple of q

Admissibility $\rightarrow$ absence of monopole?

- Magnetic monopole
 - ▶ Magnetic defect operators provide nontrivial topology
 - ▶ Quite heavy but significant in nonperturbative dynamics
 - ▶ Maxwell equation w/ monopole current j_m : $d \star F = j_e$, $dF = j_m$.
- **Admissibility** $dF = 0$ to reinstate topo. structure on lattice fields

- ▶ Exercise: $0 < \epsilon < \pi/3$?

$$F_p \equiv \frac{1}{i} \ln U_p, \quad a_\ell \equiv \frac{1}{i} \ln U_\ell \text{ in } (-\pi, \pi]; \quad F_p = (da)_p + 2\pi N_p \text{ since}$$

$$(da)_p = \Delta_\mu a_{n,\nu} - \Delta_\nu a_{n,\mu} = [a_{n+\hat{\mu},\nu} - a_{n,\nu}] - [a_{n+\hat{\nu},\mu} - a_{n,\mu}]$$

$$|da| \leq 4|a| \leq 4\pi; \quad \text{Introduce } N \text{ as } |da + 2\pi N| \leq \pi.$$

$$\text{Imposing } |F_p| < \epsilon, \quad 6\epsilon > \underbrace{|\varepsilon_{\mu\nu\rho\sigma} \Delta_\nu F_{\rho\sigma}|}_{\rightarrow 0} = 2\pi \underbrace{|\varepsilon_{\mu\nu\rho\sigma} \Delta_\nu N_{\rho\sigma}|}_{< 1};$$

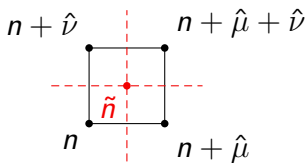
therefore $6\epsilon < 2\pi$

- How to discuss monopole properties on lattice w/ admissibility?
 - ▶ 2D compact bosons: θ term and magnetic operators
 - ▶ Observation of analogue of Witten effect
 - ▶ Future study: 4D gauge theory

Lattice formulation of θ angle and Witten effect

- Compact scalar fields: $\phi_1(n)$, $\phi_2(\tilde{n})$

- ▶ Dual lattice $\tilde{n} = n + \frac{1}{2}\hat{1} + \frac{1}{2}\hat{2}$
 - ▶ $\partial\phi_a(s, \mu) \equiv \frac{1}{i} \ln e^{-i\phi_a(s)} e^{i\phi_a(s+\hat{\mu})}$
 - ▶ $\sup_{\ell} |\partial\phi_{a,\ell}| < \epsilon$, $0 < \epsilon < \pi/2$;
then Bianchi identity $d\partial\phi = 0$



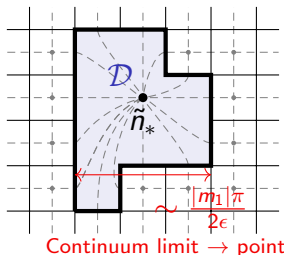
- Lattice action with θ angle

$$S = \frac{R^2}{2\pi} \sum [1 - \cos \partial\phi_{a,\ell}] - i\theta Q, \quad Q = -\frac{1}{4\pi^2} \sum \varepsilon_{\mu\nu} \partial\phi_{2,(\tilde{n},\mu)} \partial\phi_{1,(n+\hat{\mu},\nu)}$$

- Usually, $\mathcal{Q}_{m1}(C) \equiv \frac{1}{2\pi} \sum_{\ell \in C} \partial\phi_{1,\ell} = 0$
- **Excision method**: sites&links eliminated inside $\mathcal{D}$; put one dual site $\tilde{n}_*$ in $\mathcal{D}$

$$m_1 = \mathcal{Q}_{m1}(\partial\mathcal{D}) \in \mathbb{Z}$$

- $\langle M_1(\tilde{n}_*) \dots \rangle_{\theta+2\pi} = \langle M_1(\tilde{n}_*) e^{im_1\phi_2(\tilde{n}_*)} \dots \rangle_{\theta}$
for magnetic defect operator $M_1(x)$



EM symmetry and 't Hooft anomaly

- $U(1)$ electric/magnetic global symmetries: $j_e = \star d\phi$, $j_m = d\phi$
 - ▶ Electric charged object $e^{i\phi}$;
 - Monopole $M_1(x)$ with $(0, m_1) \xrightarrow{\text{Witten}}$ **dyon** with (m_1, m_1)
- Background gauging $U_{p,\tilde{p}}^{\text{em},a}$:
 $F_p^{(e,1)}$ and $F_{\tilde{p}}^{(m,1)}$ for ϕ_1 , $F_{\tilde{p}}^{(e,2)}$ and $F_p^{(m,2)}$ for ϕ_2
 - ▶ $\sup |F| < \delta$ with $0 < \delta < \min(\pi, 2\pi - 4\epsilon)$
 - ▶ $F_{\mu\nu}^e = \Delta_\mu D\phi_{n,\nu} - \Delta_\nu D\phi_{n,\mu}$
- Analogue of θ shift is given by
 $\theta \rightarrow \theta + 2\pi$, $F_{\tilde{p}}^{(m,1)} \rightarrow F_{\tilde{p}}^{(m,1)} - F_{\tilde{p}}^{(e,2)}$, $F_p^{(m,2)} \rightarrow F_p^{(m,2)} + F_p^{(e,1)}$
 - ▶ Mixing $\mathcal{Q}_{m1}$ with $\mathcal{Q}_{e2}$ on $\tilde{\Lambda}$; $\mathcal{Q}_{m2}$ with $\mathcal{Q}_{e1}$ on Λ (**Witten effect**)
- Under above shift, we observe mixed 't Hooft anomaly

$$\mathcal{Z}_{\theta+2\pi}[A^{(e,a)}, A^{(m,a)} - \varepsilon_{ab}A^{(e,b)}] = e^{-\frac{i}{2\pi} \sum \varepsilon_{\mu\nu} A_\mu^{(e,2)}(\tilde{n}) A_\nu^{(e,1)}(n+\hat{\mu})} \mathcal{Z}_\theta[A]$$

- Magnetic charge in bosonization of 2D $U(1)$ chiral gauge theory
[OM–Onoda–Suzuki '24]

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$U(1) \times U(1)'$ lattice gauge theory: Gauge

- Vector $U(1) \ni u(x, \mu)$ (physical) and Chiral $U(1)' \ni U(x, \mu)$ (non-dynamical) gauge group
- Expectation value with magnetic flux m :

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int Du e^{-S_G} \langle \mathcal{O} \rangle_F, \quad \langle \mathcal{O} \rangle_F = w[m] \int D\psi D\bar{\psi} e^{-S_F} \mathcal{O}.$$

- (Unconventional) Gauge action S_G following [Lüscher],

$$S_G = \frac{1}{4g_0^2} \sum_{x \in \Gamma} \sum_{\mu, \nu} \mathcal{L}_{\mu\nu}(x)$$
$$\mathcal{L}_{\mu\nu}(x) = \begin{cases} [f_{\mu\nu}(x)]^2 \left\{ 1 - \frac{[f_{\mu\nu}(x)]^2}{\epsilon^2} \right\}^{-1} & \text{if } |f_{\mu\nu}(x)| < \epsilon, \\ \infty & \text{otherwise.} \end{cases}$$

for $0 < \epsilon < \pi/3$, where the field strength of $u(x, \mu)$ is

$$f_{\mu\nu}(x) = \frac{1}{i} \ln u(x, \mu) u(x + \hat{\mu}, \nu) u(x + \hat{\nu}, \mu)^{-1} u(x, \nu)^{-1}.$$

$U(1) \times U(1)'$ lattice gauge theory: Fermion

- Two left-handed Weyl fermions $\psi_{1,2}$ possess $U(1)$ charges

$$(e_1, e'_1) = (+1, -1), \quad (e_2, e'_2) = (-1, -1).$$

- Since overlap Dirac operator fulfills Ginsparg–Wilson relation

$$\gamma_5 D + D \gamma_5 = D \gamma_5 D,$$

chirality projection operators are defined by

$$\hat{\gamma}_5 = \gamma_5(1 - D), \quad \hat{P}_{\pm} = \frac{1}{2}(1 \pm \hat{\gamma}_5), \quad P_{\pm} = \frac{1}{2}(1 \pm \gamma_5).$$

- Fermion action is

$$S_F = \sum_{x \in \Gamma} \bar{\psi}(x) D \psi(x) = \sum_{x \in \Gamma} \bar{\psi}(x) P_+ D \hat{P}_- \psi(x)$$

- Fermion integration measure is

$$D\psi D\bar{\psi} = \prod_j dc_j \prod_k d\bar{c}_k, \quad \psi(x) = \sum_j v_j(x) c_j, \quad \bar{\psi}(x) = \sum_k \bar{c}_k \bar{v}_k(x)$$

Basis $v_j(x)$ are in projected space $\hat{P}_- v_j = v_j$, $(v_k, v_j) = \delta_{kj}$

Construction of fermion integration measure

- Basis vectors depend on gauge field due to $\hat{P}_-$
- Under variation δ of gauge field, δv_j is not determined. . .
 - ▶ Requirements: (single-valued, gauge-inv) smooth, & local
- Under infinitesimal variations $\delta_\eta u(x, \mu) = i\eta_\mu(x)u(x, \mu)$,
 $\delta_\eta U(x, \mu) = i\eta'_\mu(x)U(x, \mu)$,

$$\delta_\eta \ln \det \bar{v} D v = (\text{term from } \delta_\eta D) + (\text{term from } \delta_\eta v).$$

Second term is called measure term (measure currents)

$$-i\mathcal{L}_\eta = \sum_j (v_j, \delta_\eta v_j) := -i \sum_{x \in \Gamma} [\eta_\mu(x) j_\mu(x) + \eta'_\mu(x) J_\mu(x)] .$$

- ▶ Consistent/covariant anomaly: $\mathcal{A}_{\text{cons}} = \frac{1}{3} \mathcal{A}_{\text{cov}}$ for Abelian GT
- ▶ To remedy this issue, we need to include counterterm [E.g., see Fujikawa–Suzuki Chap. 6]

Construction of measure currents

- For $\eta_\mu(x) = -\partial_\mu \omega(x)$, $\eta'_\mu(x) = -\partial_\mu \Omega(x)$, ($\gamma = -\frac{1}{32\pi^2}$)

$$\delta_\eta \langle \mathcal{O} \rangle_F = \langle \delta_\eta \mathcal{O} \rangle_F - 2i\gamma \sum_{x \in \Gamma} \Omega(x) \epsilon_{\mu\nu\rho\sigma} f_{\mu\nu}(x) f_{\rho\sigma}(x + \hat{\mu} + \hat{\nu}) \langle \mathcal{O} \rangle_F. \quad (\star)$$

- To prove this, measure currents fulfill the following conditions:
 - ① depend smoothly and locally on gauge fields
 - ② satisfy “integrability condition” ($[\delta_\eta, \delta_\zeta]$)
 - ③ satisfy “anomalous conservation law” ($\partial^* j, \partial^* J$)
- We find **non-local** dependence of $U(x, \mu)$; but $U(x, \mu)$ is external
- In infinite volume, one can construct the currents explicitly w.r.t.

$$u(x, \mu) = e^{i\mathfrak{a}_\mu(x)}, \quad |\mathfrak{a}_\mu(x)| \leq \pi(1 + 8\|x\|), \quad f(x) = d\mathfrak{a}(x),$$

$$U(x, \mu) = e^{i\mathfrak{A}_\mu(x)}, \quad |\mathfrak{A}_\mu(x)| \leq \pi(1 + 8\|x\|), \quad F(x) = d\mathfrak{A}(x),$$

as follows:

Backup: Construction of measure currents $L \rightarrow \infty$

$$\begin{aligned}
 \mathfrak{L}_\eta^{\star\text{inv}} &= i \int_0^1 ds \operatorname{Tr}(\hat{P}_- [\partial_s \hat{P}_-, \delta_\eta \hat{P}_-]) + \int_0^1 ds \sum_{x \in \mathbb{Z}^4} [\eta_\mu(x) k_\mu(x) + \mathfrak{a}_\mu(x) \delta_\eta k_\mu(x)] \\
 &\quad + \int_0^1 ds \sum_{x \in \mathbb{Z}^4} [\eta'_\mu(x) K_\mu(x) + \mathfrak{A}_\mu(x) \delta_\eta K_\mu(x)] \\
 &\quad - \frac{4}{3} \gamma \sum_{x \in \mathbb{Z}^4} \epsilon_{\mu\nu\rho\sigma} \{ \eta'_\mu(x) \mathfrak{a}_\nu(x + \hat{\mu}) f_{\rho\sigma}(x + \hat{\mu} + \hat{\nu}) \\
 &\quad \quad \quad + \mathfrak{A}_\mu(x) \delta_\eta [\mathfrak{a}_\nu(x + \hat{\mu}) f_{\rho\sigma}(x + \hat{\mu} + \hat{\nu})] \} \\
 &:= \sum_{x \in \mathbb{Z}^4} [\eta_\mu(x) j_\mu^{\star\text{inv}}(x) + \eta'_\mu(x) J_\mu^{\star\text{inv}}(x)] , \\
 \mathfrak{L}_\eta^{\star\text{non-inv}} &= 4\gamma \sum_{x \in \mathbb{Z}^4} \epsilon_{\mu\nu\rho\sigma} \{ \eta'_\mu(x) \mathfrak{a}_\nu(x + \hat{\mu}) f_{\rho\sigma}(x + \hat{\mu} + \hat{\nu}) \\
 &\quad \quad \quad + \mathfrak{A}_\mu(x) \delta_\eta [\mathfrak{a}_\nu(x + \hat{\mu}) f_{\rho\sigma}(x + \hat{\mu} + \hat{\nu})] \} \\
 &:= \sum_{x \in \mathbb{Z}^4} [\eta_\mu(x) j_\mu^{\star\text{non-inv}}(x) + \eta'_\mu(x) J_\mu^{\star\text{non-inv}}(x)] .
 \end{aligned}$$

- 6 Higher-group structure in lattice gauge theory
[Kan–OM–Nagoya–Wada, Abe–OM–Onoda]
 - Higher-group symmetry under modification of instanton sum
- 7 Magnetic operators and Witten effect on the lattice
[Abe–OM–Onoda–Suzuki–Tanizaki (OM–Onoda–Suzuki)]
 - Admissibility condition and monopole
 - Magnetic operators in lattice 2D scalar theory
- 8 $U(1) \times U(1)'$ lattice gauge theory with Ginsparg–Wilson fermion
 - Construction of chiral fermion integration measure [Lüscher]
- 9 “Gauge average” prescription for non-invertibility
[Honda–Onoda–OM–Suzuki]
 - Harada–Tsutsui formalism
 - Symmetry operator under gauge average

“Gauge average” prescription in anomalous GT

- Prescription by Karasik [’22] instead of [Choi–Lam–Shao ’22]
 - ▶ “Gauge average” (similar to Harada–Tsutsui formalism)

$$\tilde{U}_\alpha(M) = \int D\phi \hat{U}_\alpha(M)|_{A \rightarrow A - d\phi}$$

- What is anomalous gauge theory [Kikukawa–Suzuki]?

In other words, do we have to know an anomaly-free fundamental theory very precisely to study dynamics of chiral gauge theories at an (low) energy scale of our concern?

- ▶ [Faddeev–Shatashvili 84’, 86’, ...]
- Harada–Tsutsui formalism [’87]
 - ▶ Faddeev–Popov determinant as (g denotes a gauge transf)

$$\Delta_f[A] \int \mathcal{D}g \delta(f[A^g]) = 1,$$

- ▶ Integration of Wess–Zumino action over g
→ gauge-invariant effective action with non-local terms.
- There are some subtleties → [an other method by Tanizaki-san](#)

Gauge average and projection

- U_α is not invariant under gauge transf on **boundary variables**
- An operator defined by average over gauge transf

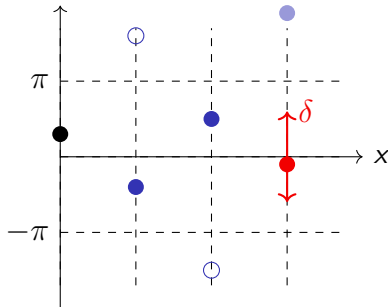
$$\langle \tilde{U}_\alpha(\tilde{\mathcal{M}}_3) \mathcal{O} \rangle_F := \langle \mathcal{O} \rangle_F^{\tilde{\mathcal{M}}_3} \int D\lambda e^{[i\alpha\gamma \sum_{x \in \mathcal{M}_4} \epsilon_{\mu\nu\rho\sigma} f_{\mu\nu}(x) f_{\rho\sigma}(x + \hat{\mu} + \hat{\nu})]^\lambda}$$

[$^\lambda$] indicates

$$u(x, \mu) \rightarrow \lambda(x) u(x, \mu) \lambda(x + \hat{\mu})^{-1}$$

- Smoothness condition

$$\frac{1}{i} \ln \lambda(x) \lambda(x + \hat{\mu})^{-1}$$



- Winding k_ν can be defined under this **gauge non-inv** condition

- $|k_\nu| < \frac{\delta}{2\pi} L$ with lattice size L ;

if $L \rightarrow \infty$,

$$\sum_{k=-\infty}^{\infty} e^{ikx} \propto \sum_{n=-\infty}^{\infty} \delta(x - 2\pi n)$$

- Gauge average implies (S_μ^1)

$$\delta \left(\frac{\alpha}{2\pi} \frac{1}{4\pi} \sum_{x \in \mathcal{M}_2^\nu} \epsilon_{\mu\nu\rho\sigma} f_{\rho\sigma}(x) - \mathbb{Z} \right)$$

- ▶ $\frac{\alpha}{2\pi}$ irrational: no magnetic flux
- ▶ $\frac{\alpha}{2\pi} = \frac{p}{N}$: $\frac{1}{2\pi} \int_{\mathcal{M}_2} da = N\mathbb{Z}$

- $\tilde{U}(\tilde{\mathcal{M}}_3) = U(\tilde{\mathcal{M}}_3) P(\tilde{\mathcal{M}}_3)$

- ▶ P for allowed magnetic fluxes

Some subtleties in Karasik prescription on lattice

- We constructed non-invertible symmetry operator $\tilde{U}$ following Karasik in lattice gauge theory
 - ▶ Non-locality of $U(x, \mu)$ in fermion integration measure
 - ▶ Gauge non-invariant constraint for gauge transf; but irrelevant in continuum limit?
 - ▶ Non-intrinsically 3D construction in reference to auxiliary 4D bulk
 - ▶ (Are relative weights correct? Really topological?)
- Instead of CS, **lattice $\mathbb{Z}_N$ TQFT** (thanks to Tanizaki-san)
 - ▶ Set rotation angles α as $2\pi p/N$ ($p, N \in \mathbb{Z}$)
 - ▶ It is natural to consider 3D level- N BF theory

$$S_{\text{BF}} = -\frac{ip\pi}{N} \sum_{x \in \mathcal{M}_3} \epsilon_{\mu\nu\rho} \left\{ b_\mu(\tilde{x}) \left[\partial_\nu c(x + \hat{\mu}) - \frac{1}{2} z_{\nu\rho}(x + \hat{\mu}) \right] - \frac{1}{2} z_{\mu\nu}(x) c_\rho(x + \hat{\mu} + \hat{\nu}) \right\}$$

- ★ Dual lattice $\tilde{x}$; $f = \delta a + 2\pi z$, $-\pi < a_\mu(x) \equiv \frac{1}{i} \ln u(x, \mu) \leq \pi$
- ▶ For simplicity, $S_{\text{BF}} = \frac{-ip\pi}{N} \sum [b(\delta c - z) - z \cup c]$

Backup: Gauge average and *smooth* gauge transf

- U_α is not invariant under gauge transf on **boundary variables**
- An operator defined by average over gauge transf

$$\begin{aligned} & \left\langle \tilde{U}_\alpha(\tilde{\mathcal{M}}_3)\mathcal{O} \right\rangle_F \\ & := \langle \mathcal{O} \rangle_F^{\tilde{\mathcal{M}}_3} \int D\lambda \exp \left[i\alpha\gamma \sum_{x \in \mathcal{M}_4} \epsilon_{\mu\nu\rho\sigma} f_{\mu\nu}(x) f_{\rho\sigma}(x + \hat{\mu} + \hat{\nu}) \right]^\lambda \end{aligned}$$

- $[\]^\lambda$ indicates gauge transf on boundary variables

$$u(x, \mu) \rightarrow \lambda(x) u(x, \mu) \lambda(x + \hat{\mu})^{-1}, \quad \lambda(x) = e^{-i\phi(x)}, \quad -\pi < \phi(x) \leq \pi.$$

- Impose smoothness (gauge non-inv) condition on possible λ :

$$\text{for } -\pi < \frac{1}{i} \ln \left[e^{-i\phi(x)} e^{i\phi(x+\hat{\mu})} \right] = \partial_\mu \phi(x) + 2\pi l_\mu(x) \leq \pi,$$

$$\sup_{x, \mu} |\partial_\mu \phi(x) + 2\pi l_\mu(x)| < \delta, \quad 0 < \delta < \pi/6.$$

$$\text{Then } f_{\mu\nu}(x) \rightarrow f_{\mu\nu}(x) + \partial_\nu \phi(x + \hat{\mu}) + 2\pi l_\nu(x + \hat{\mu}).$$

- Gauge-inv of CS term is realized by this condition on the lattice.

Backup: Sum over winding number and projection

- Assume $\widetilde{\mathcal{M}}_3 = T^3$ is perpendicular to $\hat{\mu}$; $\widetilde{\mathcal{M}}_3 = S^1 \times \widetilde{\mathcal{M}}_2^\nu$
- Introduce “scalar potential” $\varphi(x)$ as $l_\nu(x) = \partial_\nu \varphi(x)$
- Winding on a cycle (ν direction) provides additional integer k_ν to φ ; $|k_\nu| < \frac{\delta}{2\pi} L$ with lattice size L
- $\sum ff$ acquires $4\pi k_\nu \sum_{x+\hat{\mu} \in \mathcal{M}_3, x_\nu=0} \epsilon_{\mu\nu\rho\sigma} [f_{\rho\sigma}(x + \hat{\mu}) + f_{\rho\sigma}(x)]$
- Gauge average implies that

$$\int D\lambda e^{8\pi i \alpha \gamma k_\nu \sum_{x \in \mathcal{M}_2^\nu} \epsilon_{\mu\nu\rho\sigma} f_{\rho\sigma}(x)} \text{ and } \sum_{k=-\infty}^{\infty} e^{ikx} = 2\pi \sum_{n=-\infty}^{\infty} \delta(x-2\pi n),$$

and then $\delta\left(\frac{\alpha}{2\pi} \frac{1}{4\pi} \sum_{x \in \mathcal{M}_2^\nu} \epsilon_{\mu\nu\rho\sigma} f_{\rho\sigma}(x) - \mathbb{Z}\right)$ if $L \rightarrow \infty$.

- ▶ $\alpha/(2\pi)$ is irrational: no magnetic flux
- ▶ $\alpha/(2\pi)$ is rational (p/N): $\frac{1}{2\pi} \int_{\mathcal{M}_2} da = N\mathbb{Z}$
- $\tilde{U}(\widetilde{\mathcal{M}}_3) = U(\widetilde{\mathcal{M}}_3)P(\widetilde{\mathcal{M}}_3)$, P is a projection operator for allowed magnetic fluxes